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Aerosol pollution and economic growth in Thailand with a machine learning analysis of the environmental Kuznets Curve

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Abstract

Aerosol pollution is a chronic air pollution problem that occurs every year in Thailand. It is mostly caused by human activities and climate conditions, and has severe negative impacts on both human health and the economy. This study investigates the relationship between aerosol pollution and various economic and environmental factors using the Environmental Kuznets Curve (EKC) framework. Additionally, it forecasts aerosol pollution through 2050 by applying four machine learning approaches and comparing their accuracy and predictive performance. The results indicate that the Random Forest (RF) model provided the most accurate estimation across the train, validation, and test periods. The study found an inverted U-shape relationship between aerosol pollution and economic growth, consistent with an EKC pattern. This suggests that as Thailand's economy expands, aerosol pollution is projected to decline at an average annual rate of -0.0569% to -0.0636% from 2000 to 2050, except for the Southern region (R4), which shows an increasing rate. Furthermore, environmental features, specifically regional characteristics (R), Normalized Difference Vegetation Index (NDVI), rainfall, and wind speed, strongly influence aerosol pollution levels. Among economic features, population density was found to positively influence aerosol pollution. The study's findings indicate that population density and GPP per capita remain primary drivers of aerosol pollution in high-activity regions. Consequently, although not directly analyzed in our models, we propose introducing green finance instruments as a strategic mechanism to decouple economic activity from environmental degradation.

Keywords Aerosol pollution, Environmental Kuznets Curve, Machine learning, Remote sensing, Thailand

JEL classifications C38, Q53, Q54, Q56

1 Introduction

Increases in aerosol pollution concentrations, such as dust, PM10, PM2.5, and smog, have significant impacts on the global economy by increasing healthcare costs associated with air pollution exposure. These costs were estimated at approximately USD 300 billion, based on assessments of the willingness to pay to reduce global air pollution in 2015



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[40, 63]. Furthermore, it is projected that by 2060, air pollution will result in economic costs equivalent to 1% of global GDP, affecting sectors such as health, the environment, agriculture, and the global tourism industry. In West Africa, dust pollution was found to reduce economic growth by 3% over two years and to reduce agricultural yields by an average of 2% annually [17]. Additionally, a study conducted in Almería, Spain, found that dust contamination reduced the power output of a photovoltaic plant by an average of 5%, resulting in annual economic losses of more than €150,000 for the industrial facility [3]. In China, total emissions reached 11.9 billion tons in 2021, accounting for a 33% global share, and 34% of cities did not show a significant decline in PM_{2.5} concentrations [70]. In Ho Chi Minh City, Vietnam, it was found that road dust from vehicles contains heavy metal components such as lead and manganese, causing health problems in children and an increased risk of cancer in adults. In addition, a study in Hanoi showed that increases in PM₁, PM_{2.5}, and PM₁₀ concentrations were associated with 2.5%, 2.2%, and 1.4% increases, respectively, in hospitalizations for respiratory diseases [14, 58]. Finally, in India, it was found that air pollution problems caused a high cost of illness. This total loss of \$36.8 billion was equivalent to 1.36% of India's gross domestic product (GDP) [43, 49].

Thailand's insufficient attention to environmental issues has led to worsening annual air pollution, resulting in high economic costs. In 2019, environmental management agencies received just 0.4% of the national budget, or 0.05% of GDP, well below allocations in regions such as the European Union, South America, and China [6]. This chronic underfunding contributes directly to a rising toll: World Bank [65] estimated 31,173 premature deaths from air pollution in 1990, rising to 48,819 by 2013, and economic costs rising from 210,603 billion Baht to 871,300 billion Baht over that period. Attavanich, [6] found that PM_{2.5} pollution alone caused economic damage totaling 2.173 million Baht to Thai households in 2019. Each year, severe PM_{2.5} pollution peaks in winter and summer, harming transportation, public health, and the economy. The negative impacts of PM_{2.5} include: reduced labor productivity due to illness; increased costs for self-protection; exacerbated social inequality as low-income groups struggle to protect themselves; diminished quality of life and restricted outdoor activities; decreased real estate values in high-pollution areas; and increased government spending to address the problem [55].

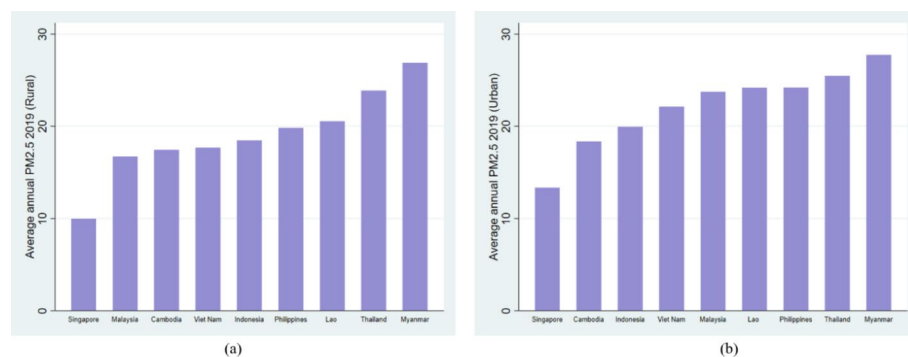


Fig. 1 **a** Average annual PM_{2.5} 2019 in Rural, **b** Average annual PM_{2.5} 2019 in Urban. Source: WHO [63] Moreover, Comparing PM_{2.5} concentrations across ASEAN countries in 2019, Thailand ranked among the top countries in the region, with both urban and rural areas showing high levels, as shown in Fig. 1.

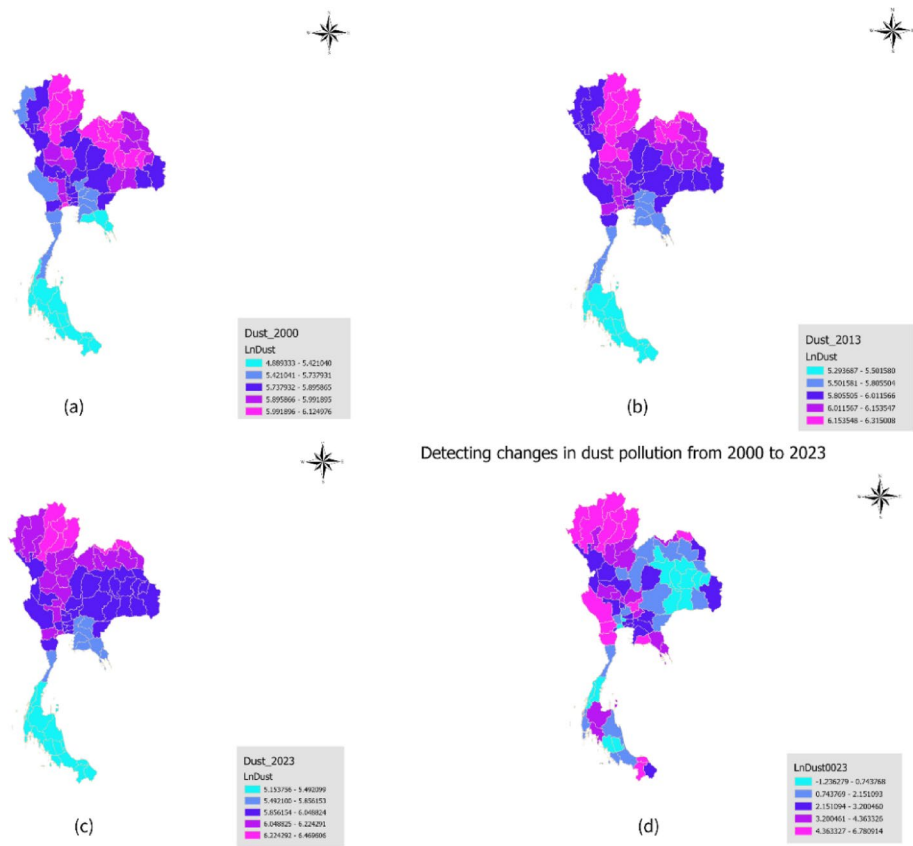


Fig. 2 **a** Average annual aerosol pollution in Thailand 2000, **b** Average annual aerosol pollution in Thailand 2013, **c** Average annual aerosol pollution in Thailand 2023, **d** Change detection of aerosol pollution in Thailand 2000–2023. Source: Author’s calculation

Table 1 Thai provinces and regions	
Regions	Provinces
Bangkok and vicinities, central, western, eastern (R1)	Bangkok Metropolis, Samut Prakan, Pathum Thani, Samut Sakhon, Nakhon Pathom, Nonthaburi, Saraburi, Singburi, Chai Nat, Ang Thong, Lop Buri, Phra Nakhon Sri Ayuthaya, Ratchaburi, Kanchanaburi, Phachuap Khiri Khan, Phetchaburi, Suphan Buri, Samut Songkhram, Chon Buri, Chachoengsao, Rayong, Trat, Chanthaburi, Nakhon Nayok, Prachin Buri, Sa Kaew
Northeastern (R2)	Khon Kaen, Udon Thani, Loei, Nong Khai, Mukdahan, Nakhon Phanom, Sakon Nakhon, Kalasin, Nakhon Ratchasima, Chaiyaphum, Yasothon, Ubon Ratchathani, Roi Et, Buri Ram, Surin, Maha Sarakham, Si Sa Ket, Nongbua Lamphu, Amnat Chareon
Northern (R3)	Chiang Mai, Lampang, Uttaradit, Mae Hong Son, Chiang Rai, Phrae, Lamphun, Nan, Phayao, Nakhon Sawan, Phitsanulok, Kam Phaeng Phet, Uthai Thani, Sukhothai, Tak, Phichit, Phetchabun
Southern (R4)	Phuket, Surat Thani, Ranong, Phangnga, Krabi, Chumphon, Nakhon Si Thammarat, Songkhla, Satun, Yala, Trang, Narathiwat, Patthalung, Pattani

Source: Authors

Additionally, we present the spatiotemporal trends of aerosol pollution across 76 Thai provinces from 2000 to 2023 (Fig. 2). In this study, we do not include Buengkan province [48] because we cannot obtain the previous dataset of this province from 2000 to 2011 (Table 1).

It is evident that aerosol pollution is severe in Thailand’s Northern, Northeastern, and Central regions. These areas are agricultural, densely populated, and industrial. They are

also affected by aerosol pollution blown from neighboring countries, which results from agricultural waste burning and wildfire, as shown in Fig. 2.

1.1 Using machine learning to predict aerosol pollution in conjunction with the EKC theory

Determining the relationship between aerosol pollution and economic growth remains a significant challenge because governments can use these findings to inform decisions on environmental policies and related matters in advance. Moreover, machine learning is a powerful tool for finding patterns in data and predicting trends with larger, more complex datasets. These tools can be used to build real-time forecasting models that require no human input. As we mentioned before, this study uses the Environmental Kuznets Curve (EKC) analysis framework to find the relationship between aerosol pollution and economic growth. This will reveal the direction of the relationship, specifically whether the aerosol pollution problem follows an inverted U or a U-shaped curve as the economy grows. The analysis is conducted using four machine learning algorithms: Linear Regression (LM), Multivariate Adaptive Regression Splines (MARS), Random Forest (RF), and Support Vector Regression (SVR). This is because machine learning can identify non-linear relationships, which can be applied alongside EKC theory as an alternative to econometric models. Furthermore, machine learning is highly effective at predicting future trends in aerosol pollution [30]. Moreover, this study employs popular interpretable machine learning methods, using Shapley Values and Partial Dependence Plots to interpret the model and explore the relationship [22].

The objectives of this study are: (1) to apply four currently popular machine learning algorithms within the Environmental Kuznets Curve (EKC) analysis framework to find the relationship between aerosol pollution and economic growth, as well as other features affecting changes in aerosol pollution in each region of Thailand; (2) to test the performance of these four machine learning algorithms in forecasting aerosol pollution and identifying future trends based on the specified model; and (3) to compare the performance of Linear Regression (LM) against other popular machine learning algorithms used in this study. The research contributions of this study include using new machine learning algorithms to forecast aerosol pollution in Thailand and utilizing provincial-level satellite data (Remote sensing) collected via the Google Earth Engine platform, as well as secondary data from government agencies. This alternative dataset is derived from satellite image processing rather than official Thai government data (e.g., aerosol pollution, wind speed, temperature, and precipitation). It also builds new knowledge regarding data analysis using machine learning algorithms in conjunction with Environmental Kuznets Curve (EKC) analysis as an alternative to using econometric models, such as in the studies of Maneejuk et al., [34], Phadkantha and Yamaka [44], Phrakhuopatnontakitti et al. [46], Shahzad and Aruga [50], Srisaringkarn and Aruga [51, 52]. The model developed in this study can then be applied to larger and more complex datasets for future environmental policy planning in Thailand. Additionally, the feature contributions identified here provide a guide for selecting variables in future advanced machine learning and econometric models. Because our machine learning approach indicates feature importance and forecasting rather than exact causal effects and magnitudes, this foundational knowledge is essential for building robust models that can yield deeper economic insights into the aerosol pollution issue. The novelty of this study lies

in being the first to analyze regional heterogeneity in aerosol pollution in Thailand using machine learning forecasting and Environmental Kuznets Curve (EKC) framework. Furthermore, it integrates environmental indicators (such as NDVI and climate) with economic drivers to provide long-term aerosol pollution projections (up to 2050) for policy planning. We analyze aerosol pollution trends using machine learning approaches (e.g., MARS, RF, and SVR) because they perform better at capturing non-linear relationships and are easier to implement. These capabilities are highly valuable for future policy planning aimed at mitigating Thailand's aerosol pollution problem. However, a notable limitation of these machine learning approaches is their inability to clearly explain the causal relationships between aerosol pollution and its contributing factors.

In the next section, we discuss the theoretical framework. The third section describes the analytical methods, and the fourth section presents the empirical results and discussion. Finally, we provide the conclusion and policy implications in the last section.

2 Literature reviews

The Kuznets Curve was originally proposed by Kuznets [31] to examine the relationship between economic growth and income inequality. Decades later, Grossman and Krueger [20] extended this concept to develop the Environmental Kuznets Curve (EKC). This theory postulates a relationship between various indicators of environmental degradation and per capita income. In the early stages of economic growth, pollution emissions increase and environmental quality declines. However, beyond a specific per capita income threshold (which varies by environmental indicator), the trend reverses: at higher income levels, further economic growth leads to environmental improvement. Consequently, the theory suggests that environmental impacts or per capita emissions follow an inverted U-shaped relationship with per capita income [20, 42, 53, 54]. Although empirical evidence supports the EKC hypothesis for certain countries, it is not universally applicable to all forms of environmental degradation, income levels, or national contexts.

The EKC relationship in this study is categorized into five patterns: (1) an increasing monotonic relationship; (2) a decreasing monotonic relationship; (3) an inverted U-shaped pattern; (4) a U-shaped pattern, which is in direct contrast to the EKC; and (5) a flat pattern, indicating that emissions are not influenced by income levels.

2.1 Air pollution policies

According to UNEP [59], there is no global legal framework for Ambient Air Quality Standards (AAQS). Similarly, a clear legal commitment to specific AAQS compatible with human life, health, and the natural environment does not yet exist in public international law. However, some key regional legal instruments on air quality—particularly in the EU—require individual signatory countries to develop relatively robust legal systems for air quality control. Currently, 66% of countries are subject to constitutional legal requirements that may, over time, be interpreted as requiring legally mandated acceptable air quality. Examples of AAQS enforcement mechanisms are shown in Table 2.

Moreover, for the technological innovation to improve environmental and air quality Hassan et al. [23] indicates that most countries must invest more in R&D to drive technical innovation and support renewable energy, thus enabling the energy transition.

Table 2 The world ambient air quality standards enforcement mechanisms

Country	Ambient air quality standard (AAQS)	Type of instrument
All EU member states	If an enforcement case is brought before the Court of Justice of the European Union (formerly the European Court of Justice, or CJEU) for the infringement of EU air quality laws, the individual EU Member State can be held accountable for failing to achieve legally mandated AAQS, and financial sanctions may be imposed as a result of persistent non-compliance.	Other primary legislation
Japan	When urgently necessary to prevent damage to human health from air pollution, the Minister of the Environment may issue necessary instructions regarding air quality to a prefectural governor or the mayor of a city (including special wards) specified by a Cabinet Order.	Policy/guidance
United States of America	Federal action can be taken against states: the US Environmental Protection Agency (EPA) can issue sanctions against a state and, if necessary, take over enforcement of the US Clean Air Act within that state.	Secondary legislation
China	Under laws that took effect in 2015, falsifying air quality monitoring data has become a crime as severe as creating the pollution itself, though there have been few high-profile prosecutions to date. However, according to a court ruling in the central city of Xian, two district environmental protection branch heads were recently imprisoned for ordering staff at national monitoring stations to falsify data.	Policy/guidance

Source: UNEP [59]

Previous research has highlighted how research spending and renewable deployment affect environmental degradation [24, 36]. However, these studies have not addressed the impact of energy transition and research spending on technological innovation. Some academics contend that more funding for R&D improves environmental quality and reduces emissions. For example, Fernández Fernández et al. [16], and Zafar et al. [68] found that technological innovation lessened pollution in industrialized countries via rebound and spillover effects. Moreover, developing renewable energy technologies helps achieve carbon neutrality [69]. In contrast, Danish et al. [13] reported that technological advances may raise CO₂ emissions.

At present, Thailand has to update air pollution control laws and environmental innovation technologies to align with the international regulation standard [47]. Consequently, Thailand urgently needs to enact more updated air pollution control act to establish better air quality standards in the future.

2.2 Application of EKC and ML

Currently, machine learning algorithms are being applied alongside the Environmental Kuznets Curve (EKC) theory to compare the estimation performance of econometric models using popular machine learning methods. The results vary depending on the data characteristics and the models used. We have compiled examples of recent research that applies machine learning algorithms to the issues of air pollution and economic growth, as shown in Table 3.

Table 3 summarizes findings from studies using machine learning algorithms, such as Random Forest and Support Vector Regression, and shows that results vary depending on data characteristics and the analytical methods used. However, an examination of the dependent (Target variable) and independent variables (Features) reveals a high degree of consistency across studies, with a common focus on the relationships between air pollution and economic variables such as GDP per capita, population density, trade openness, and urbanization, as well as environmental variables such as wind speed and

Table 3 Reviews on the relationship between economic growth and pollution from a machine learning approaches

Authors	Periods	Country/region	Methodology	Dependent variable	Independent variables	Findings
Nguyen et al. [39]	1977–2019	Vietnam	Autoregressive Distributed Lag (ARDL) and Back-Propagation Neural Networks (BPN)	CO2	GDP, Energy Consumption, Urbanization	N shaped
Jabeur et al. [25]	2007–2018	86 countries	Natural Gradient Boosting (NGBoost)	CO2	GDP per capita, Annual gross capital formation, Energy Use, Governance, Business Environment, Education, Personal Freedom	U-shaped and Inverted U-shaped
Padhan et al. [41]	1990–2019	BRICS-T	Kernel Regularized Least Squares (KRLS)	Ecological footprint indicator	Renewable energy consumption, Agriculture value added, Forest area, GDP per capita, Export diversification, Eco-innovations	Inverted U-shaped
Yao and Li [66]	1973–2021	Zimbabwe, Cameroon, Eritrea, Ghana	SHAP algorithm	Air visibility	GDP per capita, Wind direction, Wind speed, relative humidity, trade openness	Inverted N-shaped
Han and Lin [22]	1990–2018	118 countries	Artificial Neural Network (ANN)	CO2	GDP per capita, Population size, Country specific features	Inverted U-shaped
Krishnendu and Patra [30]	1980–2022	20 countries	Ridge, LASSO, Elastic Net and Kernel Regularized Least Squares (KRLS)	CO2 Emissions per capita, Internal freshwater resources per capita, Forest Cover Area per capita	GDP per capita, Population, Urban population, Agricultural output, Industrial output, Per capita energy consumption, Per capita renewable energy consumption, Trade openness, Institutional quality	Inverted U-shaped and M-shaped
Guliyev [21]	1995–2018	147 countries	Classifier-Lasso Method	Load Capacity Factor	GDP, Industrial structure, Foreign Direct Investment, Globalization Index	U-shaped and Inverted U-shaped
Boateng et al. [10]	1991–2023	West Africa	Linear regression, Random Forest, Gradient Boosting, and Support Vector Regression	GDP per capita	CO2, Renewable energy, Freshwater resources, PM25, Forest resources, Water usage, Population	U-shaped
Ali et al. [2]	1980–2022	Canada	Autoregressive Distributed Lag (ARDL) and Kernel Regularized Least Squares (KRLS)	CO2	GDP per capita, geopolitical risks, energy consumption, FDI, trade openness, financial development, ICT development	Inverted U-shaped

Source: Authors

precipitation. Therefore, this study selects aerosol pollution data obtained from the Google Earth Engine (GEE) platform as the dependent variable (Target variable), and uses economic and environmental variables such as GDP per capita, population density, NDVI, precipitation, and wind speed collected from government agencies and GEE as independent variables (Features). The study specifically focuses on the provincial level in Thailand, which addresses an existing gap. Most previous studies used country-level or continent-level data, whereas this study uses provincial-level data. This will yield findings useful for policymaking and contribute to the body of knowledge on the application of new analytical techniques in the Thai context.

The rationale for selecting these machine learning methods is their robust ability to model complex, non-linear relationships. Algorithms such as Support Vector Regression (SVR) employ kernel functions to evaluate data similarity, enabling them to identify underlying patterns and generate accurate non-linear predictions. Additionally, Multi-variate Adaptive Regression Splines (MARS) and Random Forest (RF) operate as Adaptive Basis Function Models (ABM). Because they are non-linear in their parameters, these models frequently yield significantly better predictive performance than linear approaches. Furthermore, ABM bypasses a common limitation of kernel-based methods: the inherent difficulty of specifying an optimal kernel function for complex datasets. Ultimately, these techniques build on foundational linear regression concepts while automatically capturing non-linear dynamics. This eliminates the need to manually engineer polynomial features, resulting in a more highly accurate analytical framework [38].

2.3 Stochastic impacts by regression on population, affluence, and technology model (STIRPAT)

In this study, we utilize the STIRPAT model developed by York et al., [67] as our core mathematical framework. In conjunction with the EKC theory, we use this model to examine the relationship between aerosol pollution, economic growth, and other significant variables. The STIRPAT equation is expressed as follows:

$$I_i = aP_i^b A_i^c T_i^d e_i \quad (1)$$

In this equation, the constant a act as the scaling factor. The variables b , c , and d represent the exponents of P , A , and T , respectively, which must be empirically estimated, while e denotes the error term. Specifically, P represents population, A denotes affluence (captured by $GPPpc$ and $GPPpc^2$), and T accounts for all factors influencing environmental impact beyond population and affluence (e.g., NDVI, rainfall, wind speed, daytime land surface temperature, and region-specific features). These variables are the changes in consumption and production of goods and services, and their associated environmental impacts, resulting from changes in economic variables driven by improvements in service efficiency [61, 67]. Ultimately, the STIRPAT model and EKC framework guide the empirical analysis and conclusions presented throughout this study.

3 Methodology

3.1 Model setting

In this study, we employ machine learning algorithms to forecast long-term aerosol pollution and analyze its relationship with economic growth. This analysis is grounded in the Environmental Kuznets Curve (EKC) hypothesis and utilizes the STIRPAT model as

the core mathematical framework. Feature selection is informed by the literature review presented in Sect. 2, allowing us to construct the empirical model as follows:

$$Dust = f(GPPpc, GPPpc^2, Popdens, NDVI, Rainfall, Windspeed, LST_{Day}, R) \quad (2)$$

Dust is the log of aerosol pollution in each province, *GPPpc* is the log of GPP per capita in each province, $GPPpc^2$ is its squared term, *Popdens* is the log of population density in each province, *NDVI* is the log of Normalized Difference Vegetation Index (dense and healthy green vegetation) in each province, *Rainfall* is the log of precipitation in each province, *Windspeed* is the log of windspeed in each province, *LST_{Day}* is the log of land surface temperature at daytime in each province, and *R* represents region specific features by using dummy variables. Including the squared *GPPpc* term (or $GPPpc^2$) allows the model to capture the EKC's potential turning point, where economic growth shifts from increasing to decreasing aerosol pollution. Indeed, some machine learning algorithms, such as Random Forest (RF), can capture nonlinear relationships without explicitly introducing the squared term of *GPPpc*. By introducing the squared term a priori, the algorithm can more easily detect diminishing effects, even with a limited sample size. We also consider a model without the region-specific feature to compare and demonstrate the extent to which this feature contributes to the prediction. The analytical flowchart is shown in Fig. 3. In this study, we use aerosol pollution as a proxy for dust, smog, PM2.5, and PM10, because 'aerosol pollution' is a broader technical term than these specific types of air pollution. We use aerosol pollution as the dependent variable, which has a 67% accuracy in measuring real-world ambient air conditions in the study areas [33, 35].

3.2 Data

In this study, we use data from 76 provinces in Thailand from 2000 to 2023, divided by region: BANGKOK AND VICINITIES, CENTRAL, WESTERN, EASTERN (R1), NORTHEASTERN (R2), NORTHERN (R3), and SOUTHERN (R4). The data were collected from both the GEE platform and Thai government agencies. Using remote sensing data from the GEE platform helps overcome the limitations of environmental data availability, which is otherwise restricted and does not cover all areas of Thailand. However, this study does not include Buengkan province due to limited data from 2000 to 2011.

Additionally, based on Table 4, we describe the data sources used in the study. The descriptive statistics shown are log-transformed and presented in Table 5.

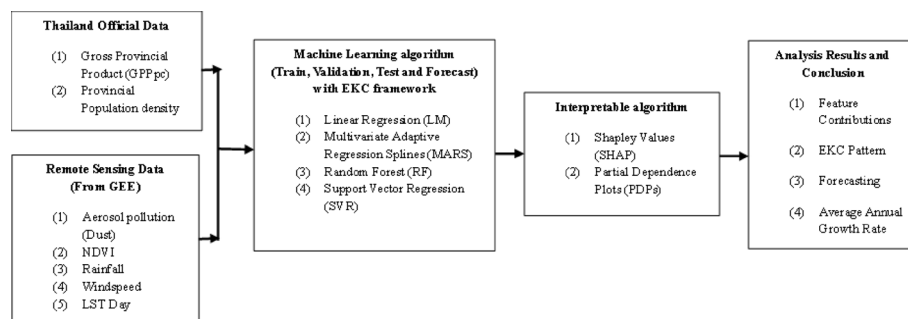


Fig. 3 Analytical flowchart. Source: Authors

Table 4 Data sources

Features	Satellite/dataset	Resolution	Frequency	Years	Units	Sources
Aerosol Pol- lution (Dust) (Optical_Depth_047)	MCD19A2.061: Terra & Aqua MAIAC Land Aero- sol Optical Depth Daily 1 km	1000 m	1 Day	2000– 2023	Max/Min	https://develo pers.google.co m/earth-engin e/datasets/cat alog/MODIS_0 61_MCD19A2 _GRANULES#d escription
Normalized Differ- ence Vegetation Index (NDVI)	MOD13A1.061 Terra Vegetation Indices 16-Day Global 500 m	500 m	16 Days	2000– 2023	Max/Min	https://develo pers.google.co m/earth-engin e/datasets/cat alog/MODIS_0 61_MOD13A1 #description
Rainfall	CHIRPS Daily: Climate Hazards Center InfraRed Precipitation With Station Data (Ver- sion 2.0 Final)	5566 m	1 Day	2000– 2023	Millimeters per day (mm/d)	https://devel opers.google. com/earth-en gine/datasets /catalog/UCS B-CHG_CHIRP S_DAILY#desc ription
Windspeed	GLDAS-2.1: Global Land Data Assimi- lation System	27,830 m	3 h	2000– 2023	Meters per second (m/s)	https://develo pers.google.co m/earth-engin e/datasets/cat alog/NASA_GL DAS_V021_NO AH_G025_T3H #description
Land Surface Tem- perature at DayTime (LST_Day)	MOD11A1.061 Terra Land Surface Temperature and Emissivity Daily Global 1 km	1000 m	1 Day	2000– 2023	Kelvin (K)	https://develo pers.google.co m/earth-engin e/datasets/cat alog/MODIS_0 61_MOD11A1 #description
Gross Provincial Product (GPPpc)	Office of the National Economic and Social Devel- opment Council (NESDC)	–	Annual	2000– 2023	Baht per Person	https://www.n esdc.go.th/inf o/gross-region al-and-provinc ial-product/
Population Density (Popdens)	Office of the National Economic and Social Devel- opment Council (NESDC)	–	Annual	2000– 2023	Person per Km ²	https://www.n esdc.go.th/inf o/gross-region al-and-provinc ial-product/

Source: Google Earth Engine and NESDC

Table 5 Descriptive statistics

	LnDust	LnGPPpc	LnGPPpc ²	LnPopdens	LnNDVI	LnRainfall	LnWindspeed	LnLST_Day
Mean	5.856	11.438	131.434	4.904	8.671	1.522	0.732	9.628
Std	0.254	0.779	18.144	0.919	0.139	0.321	0.270	0.006
Max	6.470	13.874	192.497	8.664	8.985	2.481	1.603	9.644
Min	4.889	9.636	92.851	2.769	8.137	0.682	−0.003	9.610

Source: Author's calculation

In this study, the dataset covering the interval from 2000 to 2018 (train and validation dataset) is used for model development. The subsequent period, from 2019 to 2023, is used as the test dataset to assess model performance on unseen data by fixing the year, for example, 2000–2013 for training, 2014–2018 for validation, and 2019–2023 for testing, by using a chronological split of years with the same dataset. This approach allows us to evaluate the accuracy and generalizability of our predictions. Finally, we forecasted aerosol pollution trends through 2050 based on the training and validation datasets. In other words, we build a baseline model and extend the LnDust forecast out to 2050. Excluding 2019–2023 because of the COVID-19 pandemic is the outlier data.

3.3 Machine learning techniques

In this study, four Machine Learning algorithms are used in conjunction with an interpretable approach: Shapley Values and Partial Dependence Plots.

3.3.1 Linear regression (LM)

Linear Regression, a staple of classical statistical modeling, is one of the fundamental algorithms for supervised learning. While primarily used to model linear relationships, it can be extended to model nonlinear relationships when augmented with kernels or basis function expansions. Additionally, by replacing the Gaussian output assumption with a Bernoulli or multinomial distribution, the framework can be adapted for classification tasks [38]. Fundamentally, linear regression is a statistical technique used to quantify the relationship between variables; specifically, in machine learning, it models the relationship between input features and a target label [37, 38, 62].

$$p(y|x, \theta) = N(y|w^T \varphi(x), \sigma^2) \quad (3)$$

by

$$\varphi(x) = [1, x, x^2, \dots, x^d] \quad (4)$$

or, we can explain the simple formula:

$$y = b + w_1 x_1 + w_2 x_2 + w_3 x_2^2 + \dots + w_n x_n + w_{n+1} x_{n-1}^d \quad (5)$$

y is the predicted label, the output, b is the bias of the model. Bias is the same concept as the y -intercept in the algebraic equation for a line. In Machine Learning, bias is sometimes referred to as w_0 . Bias is a parameter of the model and is calculated during training, w_1 is the weight of the feature. Weight is the same concept as the slope m in the algebraic equation for a line. Weight is a parameter of the model and is calculated during training, x_1 is a feature, the input, x_2^2 is squared term of nonlinear relationship.

Where the adaptive basis function model (ABM), which is a model of the form:

$$f(x) = w_0 + \sum_{m=1}^M w_m \varphi_m(x) \quad (6)$$

Equation (6) this adaptive basis function model (ABM) framework is applied to both Multivariate Adaptive Regression Splines (MARS) and Random Forest (RF).

3.3.2 Multivariate adaptive regression splines (MARS)

The basis equation of MARS is as follow:

$$f(x) = \beta_0 + \sum_{j=1}^D f_j(x_j) + \sum_{j,k} f_{jk}(x_j, x_k) + \sum_{j,k,l} f_{jkl}(x_j, x_k, x_l) + \dots \quad (7)$$

Where

$$f(x_1, x_2) = \beta_0 + \sum_m \beta_{1m}(x_1 - t_{1m})_+ + \sum_m \beta_{2m}(t_{2m} - x_2)_+ + \sum_m \beta_{12m}(x_1 - t_{1m})_+(t_{2m} - x_2)_+ \quad (8)$$

Multivariate Adaptive Regression Splines (MARS) offers a flexible approach for modeling nonlinear relationships by partitioning data via cutpoints, or ‘knots,’ similar to step functions [18]. The algorithm evaluates every distinct data point for each predictor as a potential knot, constructing a linear regression model using the candidate features. For instance, considering the non-linear, non-monotonic relationship $y = f(x)$. The MARS algorithm identifies the optimal split point across the range of x that minimizes error (e.g., Sum of Squared Errors) between two distinct linear segments. The resulting basis function is termed a hinge function $h(x - a)$, where a represents the cutpoint value [11]. For a single knot (Eq. (9)), the hinge function $h(x - a)$, is defined such that the piecewise linear models for y are

$$y = \begin{cases} \beta_0 + \beta_1(a - x) & x < a, \\ \beta_0 + \beta_1(x - a) & x > a \end{cases} \quad (9)$$

Once the first knot has been found, the search continues for a second knot, which is found at $x = b$ (Eq. (10)). This results in three linear models for y :

$$y = \begin{cases} \beta_0 + \beta_1(a - x) & x < a \\ \beta_0 + \beta_1(x - a) & x > a \text{ and } x < b, \\ \beta_0 + \beta_1(b - x) & x > b \end{cases} \quad (10)$$

We proceed in this way until the model becomes very large (we may impose an upper bound on the order of interactions). Then, we prune backward, at each step eliminating the basis function that causes the smallest increase in the residual error, until the cross-validation (CV) error stops improving [38].

3.3.3 Random forest (RF)

Random Forest is an ensemble learning method that aggregates predictions from multiple decision trees to yield more accurate and stable results. As a supervised learning algorithm, it is applicable to both classification and regression tasks [12]. This approach is particularly advantageous because individual decision trees often exhibit high variance and are prone to overfitting, especially when dealing with complex datasets. However, by averaging predictions across multiple trees, Random Forest effectively mitigates this variance, thereby enhancing prediction accuracy and improving the model’s overall generalization [38].

$$f_{RF}(x) = \frac{1}{M} \sum_{m=1}^M f_m(x) \quad (11)$$

$f_{RF}(x)$ is final Random Forest regression predictor, $f_m(x)$ is prediction of the m -th regression tree for input x , M is total number of trees in the forest [7].

3.3.4 Support vector regression (SVR)

Fundamentally, Support Vector Machines (SVMs) identify a separating hyperplane within an enlarged feature space, typically resulting in a nonlinear decision boundary in

the original feature space that offers robust generalization. This high-dimensional space is constructed using specific functions known as kernel functions. The principle behind Support Vector Regression (SVR) is analogous: it seeks an optimal hyperplane in the kernel-induced feature space that achieves strong generalization (measuring the similarity between objects). While various variants of SVR exist, this study employs the most prevalent formulation: ϵ -insensitive loss regression [11].

The corresponding objective function is usually written in the following form [38].

$$J = C \sum_{i=1}^N L_{\epsilon}(y_i, \hat{y}_i) + \frac{1}{2} \|w\|^2 \quad (12)$$

where $\hat{y}_i = f(X_i) = w_0 + W^T X_i$, $C = \frac{1}{\lambda}$ is a regularization constant and $\hat{w} = \sum_i \alpha_i X_i$. This objective is convex and unconstrained but not differentiable due to the absolute value function in the loss term. w is the solution vector. The epsilon-insensitive loss function, defined by

$$L_{\epsilon}(y, \hat{y}) \triangleq \begin{cases} 0 & \text{if } |y - \hat{y}| < \epsilon \\ |y - \hat{y}| - \epsilon & \text{otherwise} \end{cases} \quad (13)$$

This means that any point lying inside an ϵ -tube around the prediction is not penalized.

3.3.5 Shapley values (SHAP)

The concept of Shapley Values is grounded in cooperative game theory. It posits that the feature values of an individual observation collaborate to shift the model's prediction away from its expected base value. The method decomposes this total shift, distributing it among features based on their marginal contributions across all possible subsets, ensuring a 'fair' allocation. To do so, Shapley Values assess every combination of predictors to determine each predictor's impact. Focusing on feature x_j , the approach will test the accuracy of every combination of features not including x_j and then test how adding x_j to each combination improves the accuracy [11, 22]. The additivity property of Shapley Values is expressed in the following equation:

$$f_{SHAP}(x) = E[f(x)] + \sum_{j=1}^M \phi_j(f, x) \quad (14)$$

While $E[f(x)]$ is baseline expected value.

In the context of forecasting aerosol pollution, Shapley Values quantify the marginal contribution of predictors such as GDP per capita and regional characteristics to the model's output. Specifically, a high positive Shapley Value for GDP per capita indicates that the observed economic features drive the model to predict higher aerosol pollution relative to the baseline average. This interpretation allows us to infer the extent to which specific economic features influence environmental outcomes.

3.3.6 Partial dependence plots (PDPs)

Partial dependence plots (PDPs) illustrate the relationship between a target response and a specific set of input features by marginalizing over the values of all other input features (known as 'complement' features). Intuitively, partial dependence can be interpreted as the expected target response as a function of these input features. Given the limits of human perception, the feature set must be kept small (typically one or two dimensions);

consequently, the features of interest are usually selected from the most important ones [19].

3.3.7 Training, validation, testing and performance measurement

In this study, to tune the best machine learning algorithm for analyzing this dataset, we use the Grid Search method for hyperparameter optimization before analyzing the data with various methods. The Grid Search method tunes hyperparameters for our dataset, yielding the best values, the highest accuracy, and the lowest error. We applied Grid Search to each algorithm individually. This method is used for learning on training data, and the accuracy of each algorithm can be checked on a validation dataset, which shows the accuracy of the algorithm tuned by Grid Search. Afterward, the algorithm is truly tested on the testing dataset. Table 11 in the Appendix, in the last section, shows the best Grid Search tuning for each algorithm.

To evaluate each algorithm's performance on train, validation, and test datasets, we will use R^2 , MAE, MSE, and RMSE. The equations for these indicators are shown in Table 12 in Appendix at the last section. Table 6 lists the machine learning algorithms and their package names used in this study.

4 Results and discussions

Section 4.1 compares the performance of each machine learning algorithm, detailing how their forecasting accuracy differs. Section 4.2 presents the analysis and interpretation of the Shapley Values for each feature and explains the EKC patterns by using Partial Dependence Plots. Finally, Sect. 4.3 shows the aerosol pollution forecast until 2050 by comparing the results of each machine learning model.

4.1 Comparison of machine learning algorithm performance

Based on Table 7, which shows the train, validation, and test performance of all algorithms used to forecast aerosol pollution trends, it is evident that all algorithms demonstrate satisfactory performance. However, on the test dataset (unseen data), RF ($R^2 = 0.729$) performs well. Although the LM method ($R^2 = 0.748$) performed best during the test period, when considering R^2 , MAE, MSE, and RMSE across the train and validation periods, LM is not as effective as RF. Therefore, RF is considered the most robust algorithm for forecasting in this study, followed by SVR (0.703) and MARS (0.685). This indicates that the models used in this study can accurately forecast aerosol pollution in Thailand.

Table 6 List of R packages used in machine learning computations

Methods	Package's name	Sources
LM	lm	https://cran.r-project.org/web/packages/tidypredict/vignettes/lm.html
MARS	earth	https://cran.r-project.org/web/packages/earth/earth.pdf
RF	ranger	https://cran.r-project.org/web/packages/ranger/ranger.pdf
SVR	kernlab	https://cran.r-project.org/web/packages/kernlab/kernlab.pdf
Shapley Values	fastshap	https://cran.r-project.org/web/packages/fastshap/fastshap.pdf
Partial Dependence Plots	pdp	https://cran.r-project.org/web/packages/pdp/vignettes/pdp-intro.pdf

Source: The Comprehensive R Archive Network (CRAN)

Table 7 Model with and without region- specific features on training, validation and test results

Model with region features	Train					Validation					Test				
	MAE	MSE	RMSE	R ²		MAE	MSE	RMSE	R ²		MAE	MSE	RMSE	R ²	
Dust = fGPPpc, GPPp c ² , Popdens, NDVI, Rainfall, Windspeed, LST_Day, R															
LM	0.112	0.019	0.139	0.711		0.095	0.015	0.123	0.656		0.108	0.020	0.140	0.748	
MARS	0.100	0.016	0.126	0.765		0.103	0.018	0.133	0.597		0.125	0.024	0.156	0.685	
RF	0.098	0.015	0.123	0.776		0.078	0.010	0.101	0.769		0.114	0.021	0.145	0.729	
SVR	0.078	0.011	0.103	0.842		0.098	0.016	0.127	0.637		0.123	0.023	0.152	0.703	
Model without region features															
Dust = fGPPpc, GPPp c ² , Popdens, NDVI, Rainfall, Windspeed, LST_Day															
LM	0.124	0.025	0.158	0.627		0.117	0.023	0.151	0.481		0.124	0.025	0.159	0.674	
MARS	0.113	0.020	0.142	0.701		0.107	0.019	0.139	0.561		0.126	0.026	0.161	0.664	
RF	0.108	0.019	0.136	0.722		0.081	0.012	0.108	0.737		0.128	0.027	0.163	0.657	
SVR	0.073	0.010	0.102	0.846		0.112	0.021	0.146	0.518		0.134	0.030	0.172	0.617	

Source: Author's calculation

However, comparing the performance with and without region features, as shown in Table 7, it is clear that during testing on unseen data without the region feature model, the RF algorithm provides a high R^2 value (0.657) as high as LM (0.674). Furthermore, during train and validation, its R^2 , MAE, MSE, and RMSE values are more efficient than those of all other algorithms, including the LM algorithm. Nevertheless, compared to the region feature model, it is evident that all algorithms, including RF, perform significantly worse without it (RF with region features: $R^2 = 0.729$; RF without region features: $R^2 = 0.657$). Therefore, we can conclude that region features directly affect the accuracy of aerosol pollution forecasting. The aerosol pollution problem in Thailand varies by region, depending on topography, environmental factors (such as rainfall, wind speed, and temperature), and economic factors (such as population density and economic activities). Regarding R1, this region is typically most severely affected by aerosol pollution during the winter (January–April). Sources include heavy traffic, vehicles, industrial factories, construction, and early-season burning, combined with stagnant air. However, the annual average aerosol pollution is generally lower than in the Northern region. Notably, R1 shows no effect on aerosol pollution because this study used average annual data rather than focusing on the most severe pollution periods; this methodology may yield correlation patterns that do not align with findings from other studies [32, 45]. R2 experiences its most severe aerosol pollution from February to April. Sources originate from the burning of agricultural materials for land clearing during the sugarcane harvesting season, as well as transboundary pollution drifting from the Northern region. R3 typically faces its most severe aerosol pollution during the dry season (November–April). The main causes are open burning, agricultural waste burning, stagnant air, and transboundary haze from neighboring countries. Additionally, the region's topography, characterized by mountains and basins, traps pollutants, resulting in severe air pollution. R4 generally has better air quality than others due to the influence of sea breezes and fewer aerosol pollution sources, as the region's economy is primarily focused on the service and tourism industries [15, 56, 57]. Therefore, environmental conditions and topography are important factors that affect aerosol pollution levels in each region.

From Fig. 4, we compared the performance of all algorithms in forecasting aerosol pollution during the Testing period (2019–2023). It can be seen that the R^2 for all algorithms decreases as the forecast horizon increases. Therefore, we can conclude that if all algorithms are used to predict the next 27 years, the forecasting accuracy will gradually decrease. Thus, the new dataset should be updated and new forecasts made continuously to ensure the most accurate aerosol pollution predictions. We do not include the 2021 dataset in the R^2 prediction because during this period, the COVID-19 pandemic was an outlier that could have caused unexpected R^2 patterns.

4.2 SHAP analysis interpretation

We analyzed the feature influence on aerosol pollution using SHAP values from the RF model, as it demonstrated better performance in forecasting aerosol pollution than the other algorithms. This makes RF suitable for forecasting the complex data in this study. We calculated the Shapley Values using data from the train and validation periods (2000–2018) because the test period (2019–2023) included the COVID-19 pandemic, which could introduce outliers and reduce the interpretability of the results. Figure 5

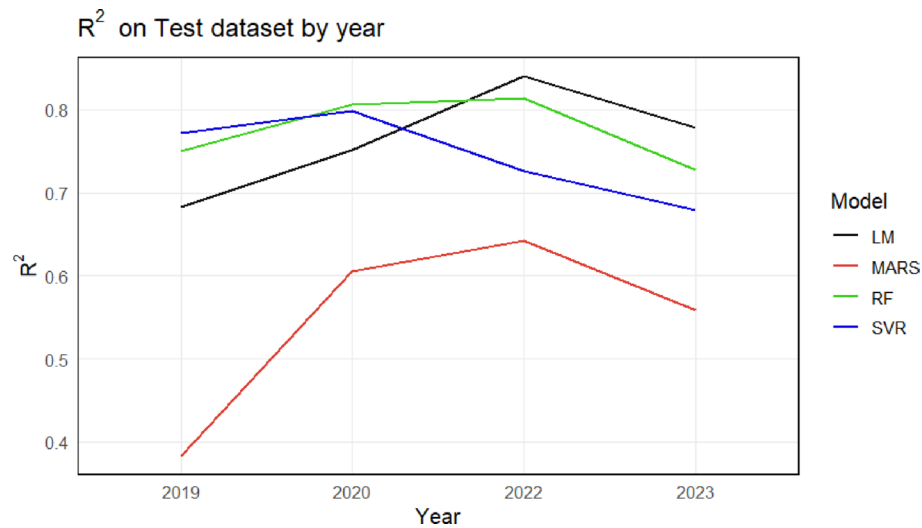


Fig. 4 R^2 comparison across models over testing period (2019–2023). We compare the R^2 performance across Machine Learning Models by using test dataset which is the unseen data. Source: Author's calculation

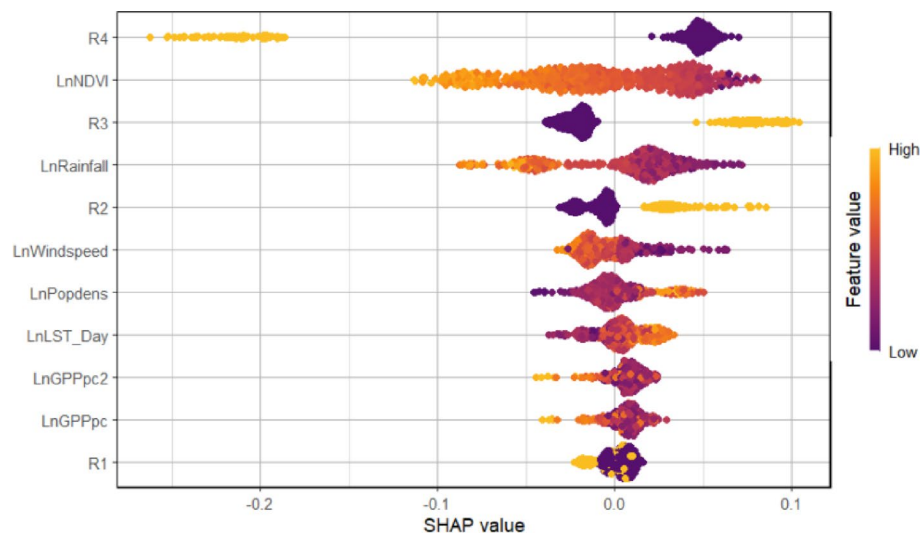


Fig. 5 SHAP summary plot for RF model which summarizes the Shapley Values from validation dataset. This figure shows the results of the SHAP summary plot. Upper features have a stronger effect on the predictive model, while lower features have a weaker effect. Source: Author's calculation

shows the SHAP summary plot for the RF model, which summarizes the Shapley Values analysis for the dataset.

4.2.1 Each feature contributions

From Fig. 5, it can be seen that the region feature (R4) has a negative influence on aerosol pollution, indicating that this region is associated with lower aerosol pollution levels. Environmental features (e.g., NDVI, rainfall, and windspeed) also negatively influence aerosol pollution; thus, areas with high values for these features (indicated in yellow) tend to have lower aerosol pollution. In contrast, Land Surface Temperature at DayTime (LST_Day) and population density both positively influence aerosol pollution. According to UNEP [60], the household sector is a major source of air pollution compared to

other sectors. However, other economic features (e.g., GPPpc and GPPpc²) show that if these features are low, aerosol pollution will be high (dark blue color). On the other hand, they show a negative influence on aerosol pollution when these features are high, meaning that if these features have high values (yellow color), aerosol pollution will be lower. This result shows an inverted U-shaped relationship between aerosol pollution and economic growth.

4.2.2 EKC analysis based on RF model

The Shapley Values analysis reveals that both LnGPPpc and LnGPPpc² have positive effects on aerosol pollution when these feature values are low, but negative effects when they are high. Consequently, the relationship between aerosol pollution and economic growth exhibits an inverted U-shaped pattern; this implies that as the Thai economy continues to grow, aerosol pollution is expected to decrease. This finding aligns with the Environmental Kuznets Curve (EKC) framework, as illustrated in Fig. 6. Figure 6a, b display the EKC patterns derived from the LM model. They clearly indicate that when LnGPPpc is low, aerosol pollution is low, but it increases as LnGPPpc rises. Conversely, when LnGPPpc² is high, aerosol pollution levels decline. Figure 6c presents the results from the RF model, which clearly demonstrates an inverted U-shaped pattern as LnGPPpc and LnGPPpc² increase. The EKC turning point is identified at an LnGPPpc of 11.492, or international \$9,335 (adjusted by Thailand PPP at 2024). This turning point value is derived using the LOESS (Locally Estimated Scatterplot Smoothing) method. LOESS is a nonparametric method for smoothing a series of data. A smooth function is a function that attempts to capture general patterns in stressor-response relationships while reducing the noise and it makes minimal assumptions about the relationship among variables.

Table 8 presents evidence for the Environmental Kuznets Curve: the base value is positive, and the slope is negative, indicating an inverted U-shaped relationship. However, R² is relatively low (0.1810 and 0.2010). It is well understood that bivariate EKC relationships often have low explanatory power because climatic factors (wind, rain,

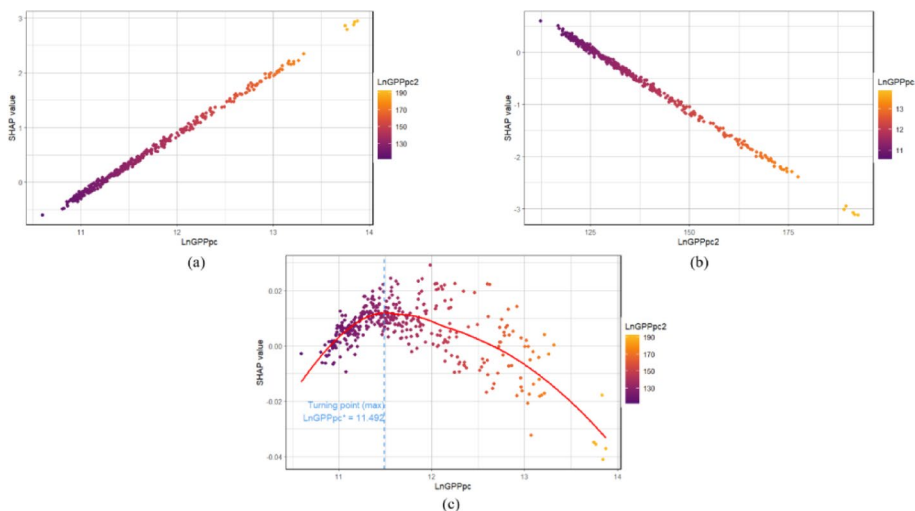
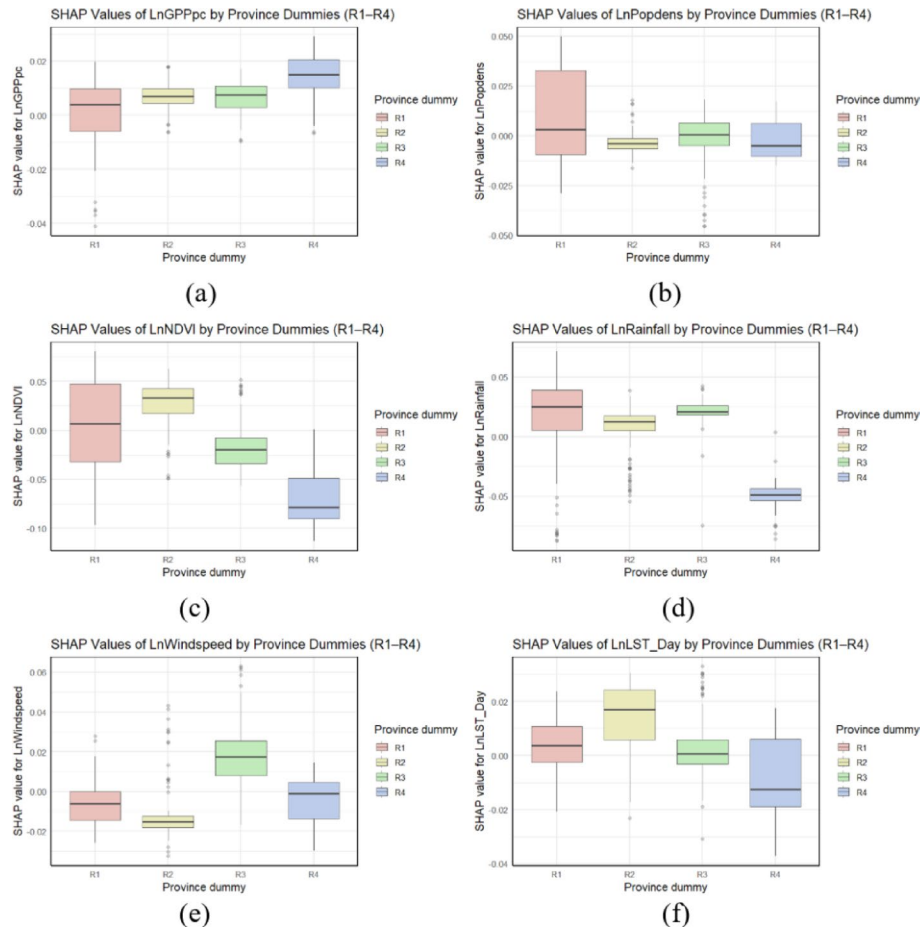


Fig. 6 Partial Dependence Plots for LM and RF models which summarizes EKC turning point: (a, b) EKC pattern from LM, (c) EKC pattern from RF. These results show an inverted U-shaped relationship with aerosol pollution (SHAP value) and economic growth. Both RF and LM results align together. Source: Author's calculation

Table 8 Evidence of the environmental Kuznets Curve statistical results

Target variable (LnDust)		Base value	Slope	R ²
Features (Input Variables)	LnGPPpc	0.0806	−0.0064	0.1810
	LnGPPpc ²	0.0442	−0.0003	0.2010

Source: Author's calculation

**Fig. 7** Shapley Values of region features by (a) GPP per capita, (b) Population density, (c) Normalized Difference Vegetation Index (NDVI), (d) Rainfall and (e) Windspeed (f) LST_Day. These figures show the magnitude of each region's impact on aerosol pollution forecasting models. Source: Author's calculation

temperature) and regional heterogeneity generate substantial variance. We also confirm the inverted U-shaped relationship from the regression analyses, as shown in Table 13 in the Appendix.

Figure 7a shows a box plot of the distribution of GPPpc relative to aerosol pollution, obtained from the Shapley Values analysis. It is evident that R1 exhibits significant variability and is a key feature determining the influence of GPPpc on aerosol pollution. Figure 7b displays the distribution of population density (Popdens); it is clear that R1 remains the primary driver of population density's influence on aerosol pollution. Figure 7c illustrates the distribution of NDVI. Here, R3 and R4 are identified as important features driving the negative influence of NDVI. This is supported by the descriptive statistics in Table 10 in the Appendix at the last section, which show that R3 and R4 have higher NDVI values than other regions. Figure 7d presents the distribution of Rainfall,

indicating that R4 is a significant factor contributing to the negative influence of Rainfall, as it experiences higher rainfall than other regions (see Table 10 in the Appendix section). Figure 7e shows the distribution of Wind Speed. It can be seen that R1 and R2 are the primary drivers of the negative influence of Wind Speed, as these regions record higher wind speeds than others (see Table 10 in the Appendix section). Finally, Fig. 7f shows the distribution of Land Surface Temperature during the daytime. It can be seen that R1 and R2 are the primary drivers of LST_Day's positive influence.

From the summary above, it is clear that we have explained the results of the Shapley Values analysis, which identify the sources of influence for each regional feature and show that region-specific factors clearly affect the aerosol pollution forecast.

4.3 Forecasting long run aerosol pollution (until 2050)

We forecast aerosol pollution through 2050 using a train and validation dataset from 2000 to 2018. We selected 2050 as the endpoint because Thailand aims to achieve net-zero greenhouse gas (GHG) emissions by then. This involves supporting green finance, promoting the use of low-emission vehicles, implementing carbon credit mechanisms, and adopting alternative energy sources (KASIKORN RESEARCH CENTER, [26]). Therefore, we used 2050 as the forecast limit. However, since forecasting accuracy naturally decreases over such a long period, new data analysis should be performed continuously to ensure the forecast remains as accurate as possible.

The results of the aerosol pollution forecast are shown in Fig. 8, broken down by regions R1–R4. R1 is the region with the highest economic activity, high urbanization, and large industrial areas. R2, R3, and R4 are mostly agricultural areas with some industrial factories, and also focus on the tourism and service sectors. Therefore, economic activity in these regions is lower than in R1. As shown in Fig. 8, the RF model has the highest accuracy in forecasting, performing very close to the LM model and other models (e.g., MARS and SVR). It should be noted that the RF model is not predicting novel “out-of-bounds” low-dust values

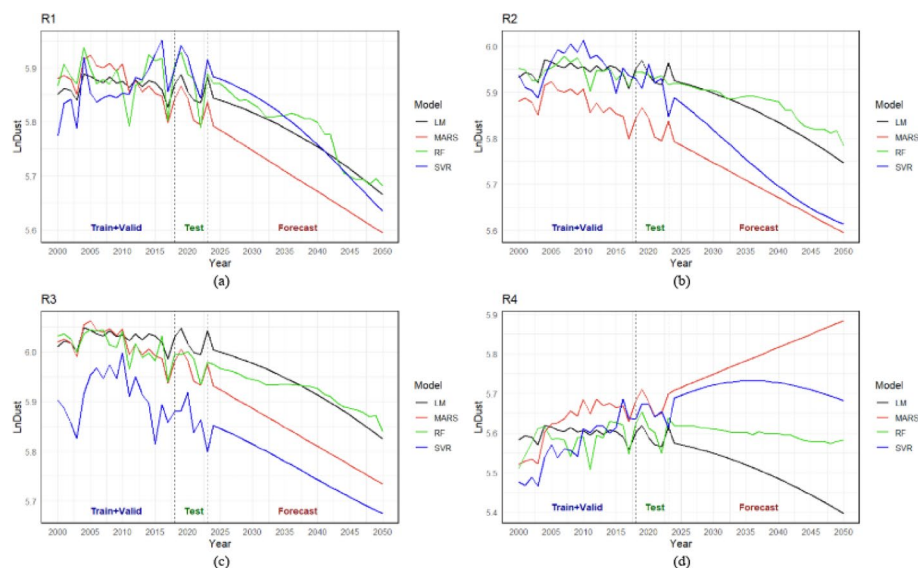


Fig. 8 Log value of Dust predictions across region features. **a** Dust prediction of R1 region (BANGKOK AND VICINITIES, CENTRAL, WESTERN, EASTERN), **(b)** R2 region (NORTHEASTERN), **(c)** R3 region (NORTHERN) and **(d)** R4 region (SOUTHERN). These figures show the predicted trend in aerosol pollution (Dust) based on economic-aerosol pollution models from 2024 to 2050. Source: Author's calculation

for the future; rather, it forecasts that future extrapolated variables will consistently route to the leaf nodes containing the lowest historically observed dust values. Because the other independent models (such as MARS and LM, which can extrapolate linearly) show a similar downward trajectory, we have confidence that the RF trend is robust and not merely an artifact of its terminal leaf nodes. However, as shown in Table 7 (with region features), RF outperforms all models, as evidenced by the R^2 , MAE, MSE, and RMSE values across train, validation, and test periods. Nevertheless, the forecast for aerosol pollution in R4 appears to be the most volatile. This is because the number of observations for the R4 region is less than for other regions, which may lead to greater forecast volatility. This is because RF is suitable for a moderate sample size, skewed, and noisy data. While other algorithms are quite sensitive to sample size and noisy data, leading to overestimation [9, 38].

From Table 9, it can be seen that the annual growth rate of aerosol pollution by region from 2000 to 2050 has a diminished downward trend in most regions. Starting with R2, it decreases by -0.0569% to -0.0990% , with an approximate average of -0.0826% per year, making it the region where aerosol pollution is trending down the most. This is followed by R3, which decreases by an average of -0.0749% per year, and by R1, which decreases by an average of -0.0679% per year. In contrast, the R4 region shows an upward trend, averaging 0.0409% per year. It is evident that the RF model produces an average annual growth rate close to the overall average across all algorithms, demonstrating its high efficiency relative to other algorithms. In summary, while R4 shows a slight increase, regions R1–R3 show a diminished downward trend in aerosol pollution continuing until 2050.

5 Conclusions and policy implications

This study investigated the relationship between aerosol pollution on economic and environmental features using the Environmental Kuznets Curve (EKC) framework. It also forecasted aerosol pollution through 2050 to identify future trends, applying four machine learning approaches and comparing their predictive accuracy and performance.

The study found that the RF model provided the most accurate estimation throughout the train, validation, and test periods compared to other machine learning algorithms. As detailed in Table 11 of the Appendix, the Random Forest (RF) grid search was configured with a minimum terminal node (leaf) size of 1, a sample fraction of 0.8, and bootstrap sampling disabled (False). Consequently, the algorithm learned from the data until each leaf contained only a single observation, permitting the trees to grow to their maximum depth. The 0.8 sample fraction split the training dataset into 80% for each individual tree, reserving the remaining 20% for error evaluation. Furthermore, disabling bootstrap sampling ensured that the algorithm did not draw duplicate data points during tree construction. Finally, the number of predictors evaluated at each split was restricted to 2, meaning the algorithm considered only two randomly chosen features to determine the optimal branching path. Because of these characteristics, RF can seamlessly accommodate mixed discrete and continuous

Table 9 Average annual growth rate of aerosol pollution by region features from 2000 to 2050

Regions	LM	MARS	RF	SVR	Average
R1	−0.0639	−0.0990	−0.0616	−0.0469	−0.0679
R2	−0.0630	−0.0990	−0.0569	−0.1113	−0.0826
R3	−0.0622	−0.0966	−0.0636	−0.0770	−0.0749
R4	−0.0670	0.1277	0.0279	0.0748	0.0409

Source: Author's calculation

inputs, perform automatic variable selection, and remain highly robust to outliers [10, 27, 28, 64]. In contrast, the other evaluated models exhibited specific limitations. Linear Regression inherently fails to capture non-linear interactions. Support Vector Regression (SVR) proved highly sensitive to scaling because its regularization parameter ($C = 4$) determines the penalty for data points falling outside the epsilon tube. While SVR ignores small errors within this threshold, it remains susceptible to outliers depending on the specific C value. Additionally, the SVR model used an RBF kernel with a sigma (σ) of 0.109, increasing the risk of overfitting to this dataset. Lastly, the Multivariate Adaptive Regression Splines (MARS) model was restricted by a maximum interaction degree of 2. Because it was strictly limited to two-way interactions, the MARS model could not effectively capture the complex, multivariable dynamic driving forces behind aerosol pollution levels [9, 38]. Thus, RF is the best model in terms of both accuracy and forecasting performance in this context. Additional studies should be conducted to affirm our findings. Furthermore, environmental features such as region-specific conditions (R), NDVI, rainfall, and wind speed strongly influence changes in aerosol pollution. Meanwhile, economic feature such as population density positively influences aerosol pollution.

This study identifies an inverted U-shaped relationship consistent with the Environmental Kuznets Curve (EKC) align with [1]. By employing the Random Forest (RF) machine learning algorithm to detect these patterns, this study offers a supportive conclusion aligned with econometric models (e.g., Linear Regression (LM)). This means that as Thailand's economy grows, aerosol pollution levels will gradually decrease. According to Fig. 6c, the EKC turning point is at 11.492 or international \$9335 (adjusted by Thailand PPP at 2024) (also confirm by regression analyses in Table 13 in the Appendix). Certainly, economic factors are clearly causes of aerosol pollution [60], including urbanization, population density, and economic growth [71]. However, because aerosol pollution is highly sensitive to environmental conditions such as wind speed, rainfall, and topography, which can reduce aerosol or blow it elsewhere, some areas in Thailand experience more severe aerosol pollution than others. As illustrated in Fig. 5, region-specific features have the most impact on aerosol pollution levels. This implies that aerosol pollution is a transboundary problem that requires urgent regional collaboration. The central government should collaborate with local governments in each region to address aerosol pollution through measures such as preventing the improper burning of agricultural materials and controlling wildfires during the dry season. Although aerosol pollution is declining, the average annual rate of decline is marginal (less than 1%). Consequently, it can be concluded that aerosol pollution will remain a persistent issue for Thailand without appropriate mitigation strategies, with the potential to intensify in the future.

5.1 Policy implications

Recognizing RF as a robust methodological instrument for assessing the influence of aerosol pollution on economic growth within the EKC framework and for constructing forecasting models is advisable. The government may draw on the empirical insights generated by this study to develop economic–aerosol pollution forecasting systems or to employ the model's feature contribution metrics to evaluate prospective causal impacts on economic performance. In this analytical context, the RF model demonstrates superior efficiency and methodological suitability.

Furthermore, the analyses showed that aerosol pollution exhibits a slightly downward trend. However, the government should plan long-term prevention measures to minimize the severity of future aerosol pollution and ensure sustainability. Examples include controlling industrial areas that cause environmental problems to minimize their impact on the public, such as zoning industrial areas according to wind direction and topography so that aerosol pollution is carried away from the public. This is because regional characteristics (R) and environmental condition features—specifically NDVI, wind speed, precipitation, and Land Surface Temperature at Daytime—have the greatest impact on aerosol pollution. Therefore, the government must keep these factors in mind when establishing industrial zones. To protect surrounding communities, these zones should be established in spacious areas with adequate air circulation and sound environmental conditions. Additional approaches to address these region-specific factors include preventing the inappropriate burning of agricultural waste and improving wildfire management, as these activities contribute significantly to transboundary pollution in neighboring regions. The study's findings indicate that population density and GPP per capita remain primary drivers of aerosol pollution in high-activity regions. Consequently, although not directly analyzed in our models, we propose introducing green finance instruments as a strategic mechanism to decouple economic activity from environmental degradation [5]. Such instruments could provide the necessary capital for manufacturing industries and business firms to adopt cleaner technologies, thereby addressing the persistent pollution issues identified in the long-run forecast through 2050. Moreover, support should be given to the enactment of a Climate Change Act and a Clean Air Act in Thailand. This serves as an integrated solution to air pollution—including transboundary pollution—and promotes stricter environmental legislation in the country.

5.2 Limitations of the study

This study has four limitations. Firstly, it did not include other important factors such as industrial density, environmental pollution control laws in Thailand, technological advancement, transboundary transport, wildfire management, agricultural burning, etc. This may limit the scope of policy recommendations and reduce the accuracy of aerosol pollution forecasting. As indicated in previous studies [8], these variables directly affect variations in aerosol pollution across regions. However, since this study utilizes provincial-level data in Thailand, the aforementioned variables could not be incorporated into the analysis. Secondly, this study used provincial-level data of Thailand, which may lead to inaccuracies in future aerosol pollution forecasts. Therefore, using more granular data, such as at the sub-district level, would make the forecast more accurate. Thirdly, if the scope of the study could be expanded, for example, by comparing with neighboring countries, it would lead to better policy recommendations and would also increase the number of observations used to train the machine learning algorithm. Fourthly, this study uses annual-average aerosol pollution data, which are influenced by topography and environmental conditions (such as wind speed, rainfall, etc.) throughout the year. Therefore, if we use seasonal data of aerosol pollution, we will see the apparent pattern of aerosol pollution in Thailand because we can control the environmental conditions that are not consistent for the whole year, but focus only on the timing of the severe aerosol pollution period [4, 29]. Moreover, while machine learning approaches cannot report statistically significant results for each feature used in the model, this limitation underscores the importance of using machine learning alongside

an econometric model (as shown in Table 13 in the Appendix) or another machine learning approach, such as Causal Forests.

5.3 Future studies

For the next study, researchers should focus specifically on the district or sub-district level, given differences in topography and environmental factors. This will make the study results more accurate. Furthermore, the use of spatial econometrics remains important, as it enables the quantification of the spatial effects of aerosol pollution. Based on our analysis, we recommend incorporating our findings into advanced machine learning (e.g., Causal Forests) and econometric models of the Thai economy and aerosol pollution, as they are significant drivers of aerosol pollution levels in Thailand.

Appendix

See Tables 10, 11, 12 and 13.

Table 10 Descriptive statistics by region-specific

Variables	Statistics	R1	R2	R3	R4
LnDust	Mean	5.896	5.952	6.026	5.443
	Std	0.181	0.111	0.163	0.146
	Max	6.313	6.270	6.470	5.822
	Min	5.335	5.638	5.541	4.889
LnGPPpc	Mean	12.036	10.805	11.161	11.523
	Std	0.766	0.513	0.484	0.510
	Max	13.874	11.834	12.374	12.954
	Min	10.366	9.636	10.152	10.250
LnGPPpc ²	Mean	145.445	117.015	124.801	133.035
	Std	18.584	11.013	10.772	11.817
	Max	192.497	140.044	153.121	167.815
	Min	107.445	92.851	103.063	105.055
LnPopdens	Mean	5.482	4.779	4.181	4.878
	Std	1.148	0.308	0.496	0.665
	Max	8.664	5.703	4.930	7.012
	Min	3.654	3.922	2.769	3.935
LnNDVI	Mean	8.626	8.568	8.716	8.840
	Std	0.133	0.086	0.080	0.078
	Max	8.860	8.845	8.869	8.985
	Min	8.137	8.360	8.470	8.585
LnRainfall	Mean	1.426	1.461	1.342	2.002
	Std	0.262	0.212	0.174	0.209
	Max	2.351	2.236	1.753	2.481
	Min	0.682	0.935	0.823	1.481
LnWindspeed	Mean	0.782	0.890	0.459	0.758
	Std	0.204	0.222	0.187	0.285
	Max	1.272	1.418	0.947	1.603
	Min	0.355	0.231	−0.003	0.190
LnLST_Day	Mean	9.630	9.632	9.625	9.623
	Std	0.004	0.004	0.007	0.003
	Max	9.644	9.642	9.644	9.632
	Min	9.619	9.617	9.610	9.616
Obs		624	456	408	336
Total Obs		1824			

Source: Author's calculation

Table 11 Optimal hyperparameter (Grid Search)

Parameters	Optimal hyperparameter	
	Model with region features	Model without region features
MARS		
Maximum degree of interaction(degree)	2	3
Maximum number of terms(nprune)	23	23
Algorithm	Caret	Caret
RF		
Number of predictors (features)	2	1
Minimum number of observations in a terminal node (leaf)	1	1
Bootstrap sampling	FALSE	FALSE
Sample Fraction	0.8	0.8
SVR		
C (regularization parameter)	4	8
Sigma	0.109137	0.2202771
Kernel	RBF	RBF

Source: Author's calculation

Table 12 Evaluation metrics

Metrics	Equation	Description
MAE	$MAE = \frac{1}{n} \sum_{i=1}^1 y_i - \hat{y}_i $	MAE offers an intuitive measure of average prediction error magnitude, presenting a clear and direct interpretation of model accuracy.
MSE	$MSE = \frac{1}{n} \sum_{i=1}^1 (y_i - \hat{y}_i)^2$	MSE extends this by squaring the errors before averaging, placing greater emphasis on larger errors, making it sensitive to outliers.
RMSE	$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^1 (y_i - \hat{y}_i)^2}$	RMSE, the square root of MSE, scales the errors back to their original units, complementing MAE by offering a view of error distribution.
R ²	$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$	R ² measures the variance in Aerosol pollution that is predictable from independent variables like GDP, population density and region features

Source: Authors

Models 2 and 3 of the LM demonstrate a statistically significant, inverted U-shaped relationship between LnDust pollution and economic growth (represented by LnGPPpc and LnGPPpc²). The FE specifications (Models 4 and 5) also indicate an inverted U-shaped relationship, although this is only statistically significant in Model 5. Similarly, the RE specifications (Models 6, 7, and 8) display an inverted U-shaped relationship, with Models 7 and 8 yielding statistically significant results. The Hausman test rejects the null hypothesis (H0), indicating that the FE model is the more appropriate specification

Table 13 Regression analyses and robustness checks

Variables/Models	LM			FE			RE		
	1	2	3	4	5	6	7	8	
Dependent Variable: LnDust									
LnGPPpc	-0.345 (-2.11)	** (8.29)	*** (11.22)	*** (1.77)	0.212 (2.53)	** (1.97)	** (3.35)	*** (4.17)	***
LnGPPpc ²	0.013 (1.91)	* (-8.13)	*** (-11.13)	*** (-1.32)	-0.011 (-1.85)	* (-1.53)	-0.014 (-2.63)	*** (-3.49)	***
LnPopdens		-0.046 (-7.76)	*** (3.36)	*** (0.021)	0.011 (0.50)		-0.039 (-2.93)	*** (0.42)	
LnNDVI		-1.150 (-24.03)	*** (-10.18)	*** (-0.522)	-0.574 (-6.40)	***	-0.782 (-10.39)	*** (-0.521)	***
LnRainfall		-0.345 (-24.23)	*** (-10.16)	*** (-0.159)	-0.157 (-8.27)	***	-0.202 (-11.34)	*** (-0.157)	***
LnWindspeed		-0.257 (-15.35)	*** (-7.27)	*** (-0.123)	-0.166 (-8.86)	***	-0.175 (-9.58)	*** (-0.160)	***
LnLST_Day		-1.831 (-1.95)	* (-0.110)		3.509 (3.00)	***	2.445 (2.17)	** (3.318)	***
R1-R4	No	No	Yes	No	Yes	No	No	Yes	
Cons	8.040 (8.47)	*** (3.18)	*** (0.67)	4.353 (6.49)	-24.997 (-2.16)	** (6.43)	*** (-1.17)	-24.603 (-2.27)	**
Obs	1,824	1,824	1,824	1,824	1,824	1,824	1,824	1,824	
R-squared	0.012	0.634	0.718	0.011	0.399	0.012	0.550	0.694	
Adj R-squared	0.011	0.633	0.716						
F-test	11.23	449.14	460.6	34.72	43.66				
Chi-squared									
Hausman Test				63.26			470.95	704.41	

Significant level: * 0.10, ** 0.05, *** 0.01

(1) LM: $\text{LnDust}_{it} = \text{LnGPPpc}_{it} + \text{LnGPPpc}_{it}^2 + \text{LnPopdens}_{it} + \text{LnNDVI}_{it} + \text{LnRainfall}_{it} + \text{LnWindspeed}_{it} + \text{LnLST_Day}_{it} + R_{1-4} + \epsilon_{it}$

(2) RE: $\text{LnDust}_{it} = \text{LnGPPpc}_{it} + \text{LnGPPpc}_{it}^2 + \text{LnPopdens}_{it} + \text{LnNDVI}_{it} + \text{LnRainfall}_{it} + \text{LnWindspeed}_{it} + \text{LnLST_Day}_{it} + R_{1-4} + \alpha_i + \epsilon_{it}(3)$

FE: $\text{LnDust}_{it} = \text{LnGPPpc}_{it} + \text{LnGPPpc}_{it}^2 + \text{LnPopdens}_{it} + \text{LnNDVI}_{it} + \text{LnRainfall}_{it} + \text{LnWindspeed}_{it} + \text{LnLST_Day}_{it} + R_{1-4} + \mu_i + \epsilon_{it}$. LM = linear regression model; FE = fixed effects model; RE = random effects model. For the LM (Models 1–3) and RE (Models 4–8) specifications, the models represent the baseline model, the model with control variables but no region-specific effects, and the model with both control variables and region-specific effects, respectively. For the FE specification, Models 4 and 5 represent the baseline model and the model with control variables and region-specific effects, respectively. Source: Author's calculations

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Author contributions

Thanakhom Srisaringkarn: Writing—original draft, Visualization, Validation, Data curation, Methodology, Conceptualization. Kentaka Aruga: Validation, Supervision, Writing—review and editing, Methodology, Funding Acquisition, Conceptualization.

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Data availability

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Declarations**Ethics approval and consent to participate**

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